7ELEVEN

# SMART DASHBOARD FOR 7ELEVEN

Team: InnoSky



7ELEVEN

# TRACK SELECTION: INTELLIGENT RETAIL ECOSYSTEMS: OPTIMIZING SUPPLY CHAINS AND OPERATIONS VIA SMART TECHNOLOGY

# COMPANY FOCUS: 7ELEVEN BANGKOK

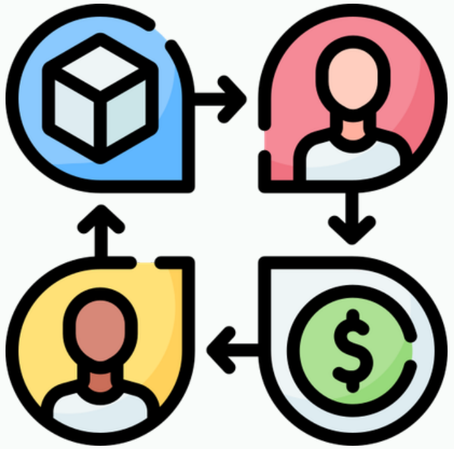





# KEY RETAIL CHALLENGES AND OPPORTUNITIES



Financial Losses



Operational Inefficiency

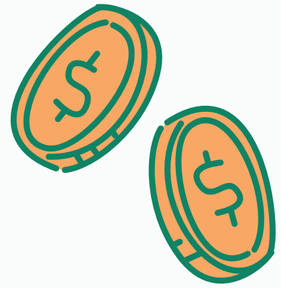


Environmental Impact

# TARGET MARKET

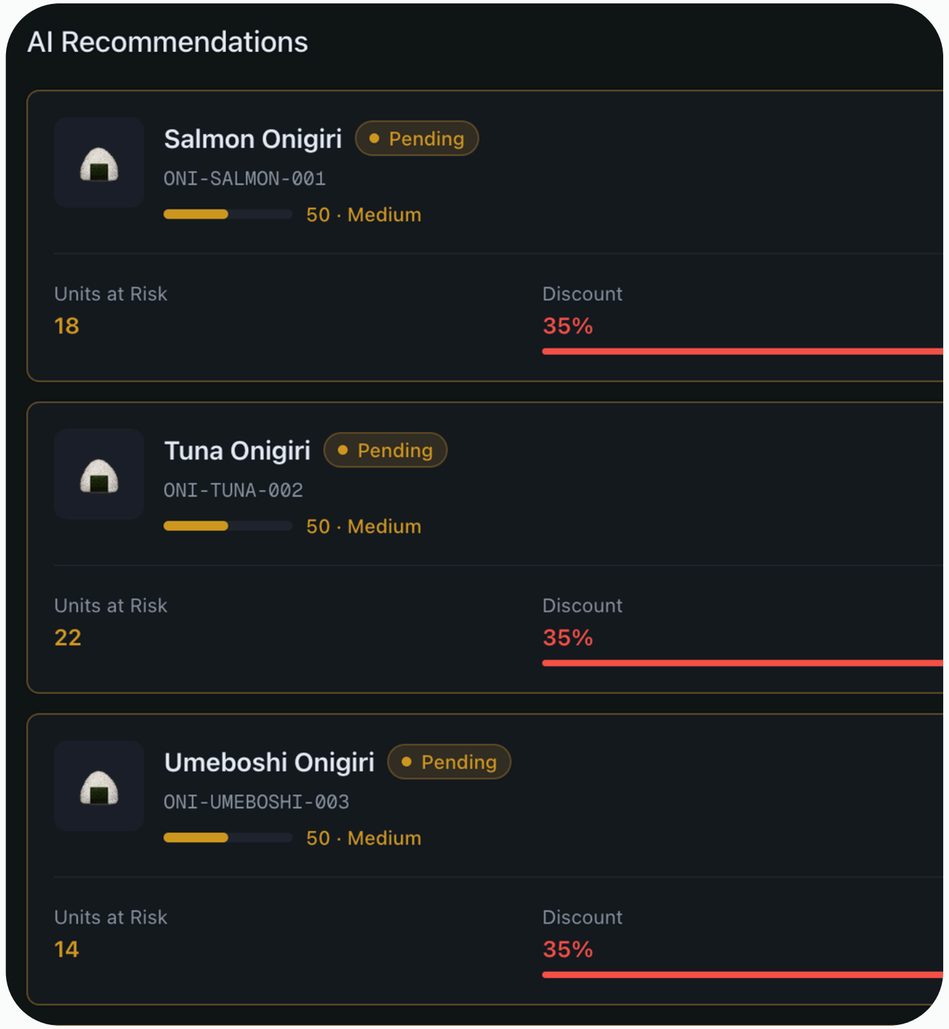
- 7Eleven Stakeholders
- 7Eleven Store Branches
- Customers





# PROBLEM STATEMENT

**7-Eleven Thailand operations team** need to make faster, data-driven discount and inventory decisions to **increase revenue** because reactive handling of near-expiry stock, volatile demand, and generic promotions leads to food waste, lost sales, and reduced revenue.






# OUR SOLUTION

**AI-powered Monitoring Dashboard for Store Inventory**

# FEATURES



| Features | Potential Impact | Status |
|---|---|---|
| Discount recommendations for slow moving and near-expiry items | • Higher sell-through, lower shrink<br>• Automates markdown decisions so stores clear risky inventory earlier, reducing expiry losses and freeing shelf space for fast movers. | Implemented |
| Branch-trait dashboards (no-alcohol zones, tourist zones) | • More relevant promos, fewer compliance issues<br>• Store-specific rules prevent “wrong promos” (e.g., alcohol near sensitive zones) and tailor bundles to local demand patterns, improving conversion while reducing manual checks. | To be Implemented |
| Weather-based demand forecasting | • Protects perishables before foot traffic drops<br>• e.g. If rain is forecasted, early markdowns and targeted promos shift perishable sales before walk-ins decline, cutting waste and stabilising daily sales volatility. | To be Implemented |

# FEATURES



| Features | Potential Impact | Status |
|---|---|---|
| Event-ending flash sales | • Captures time-window demand spikes<br>• Linking sales drops to event (concerts) end times converts “leaving the venue” crowds into immediate purchases (snacks, drinks, ready-to-eat), increasing basket size during predictable surges. | To be Implemented |
| Expiry-linked discount floors | • Control and franchise fairness<br>• Guardrails prevent over-discounting/abuse, protect margin, and still allow franchisees flexibility within safe ranges—better governance with less HQ policing. | To be Implemented |

# FEATURES

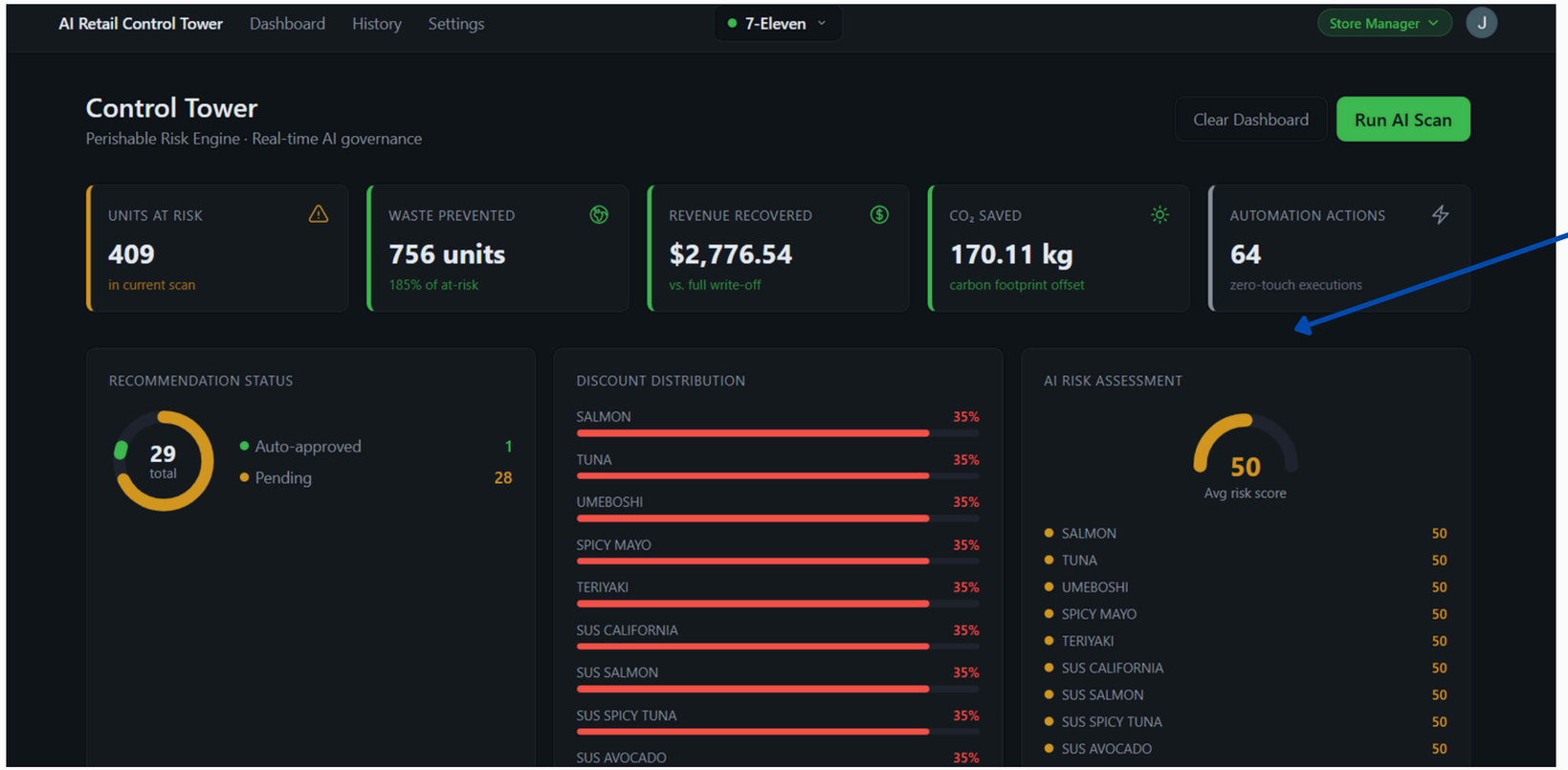


| Features | Potential Impact | Status |
|---|---|---|
| “Surprise Flash Sales” targeted to nearby customers when a store has promotions or high near-expiry or slow-moving stock. | • Fast stock clearance and stronger local competitive edge<br>• Geo-targeted deals motivate nearby customers to buy quickly, converting near-expiry stock into sales and standing out against close-by competitors in a price-sensitive market. | To be Implemented |
| Personalised push notifications (purchase history + proximity) | • Higher redemption, less promo waste<br>• Send in-app promo notifications based on what customers buy and when they are nearby, driving higher conversions than generic mass campaigns | To be Implemented |

# PROTOTYPE DEMO

Website: https://retail-sentinel.vercel.app/

Status: Actively developing




**Dashboard for tracking key metrics**

A centralized dashboard that monitors the most important operational and business metrics, allowing users to quickly assess performance and make informed decisions.

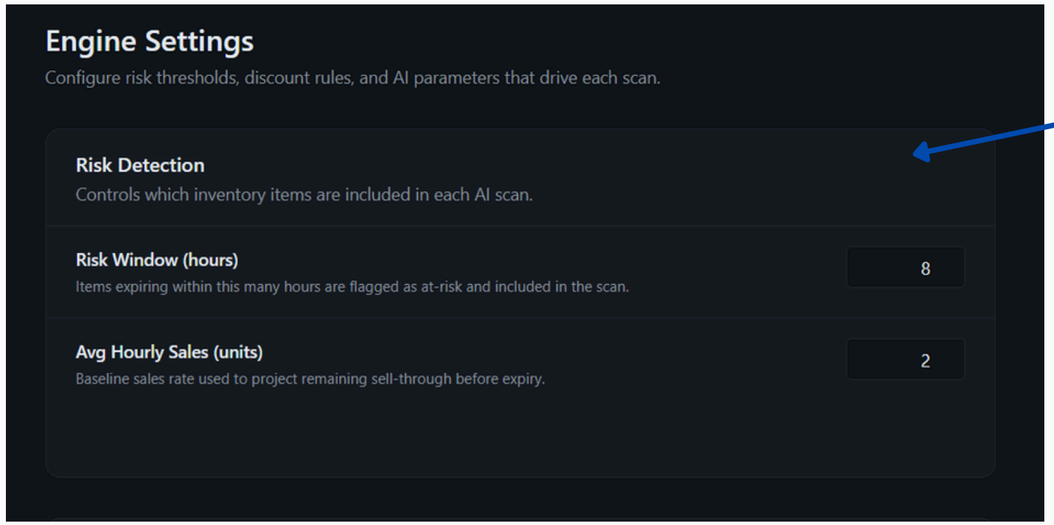
# PROTOTYPE DEMO




**AI-powered discount recommendation and approval**

System analyzes each at-risk item and suggests a discount based on expiry urgency and sales conditions. The final recommendation is still subject to user review (“allow” or “deny”), allowing managers to approve or reject the proposed discount before execution.

# PROTOTYPE DEMO

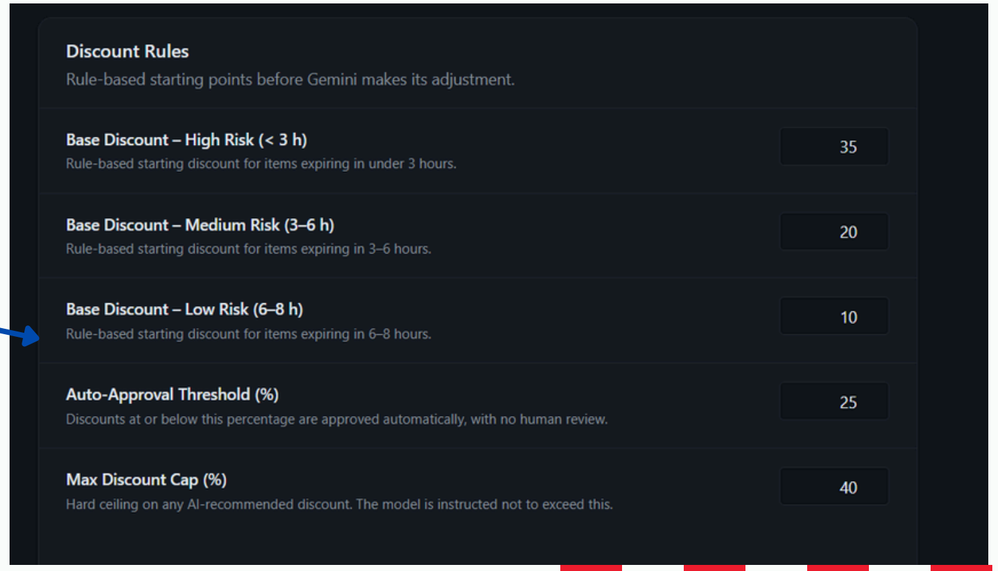



**Configurable threshold limits for the risk period**

For example, users can set the expiry risk window that determines when an item will be flagged for action.

**Configurable discount thresholds and maximum caps**

Users can set discount levels based on expiry risk and define the upper limit for discount.



# OVERALL IMPACT

Pareto principle:
20% of system adminstration tasks are responsible for 80% of your workload

- Reduce manual admin work (discounts, bundles)
- Identify slow-moving and near-expiry items early
- Be reactive: Trigger actions proactively using weather and event forecasts
- Reduce waste with timely markdowns on perishables
- Protect margins with pricing rules and minimum price floors
- Increase sales with branch-specific “Quick-Move Bundles”
- Standardise decisions across branches and reduce human errors



# Thank you!